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Report

Analyzing Retail Sales Data

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Problem statement

The retail has difficulty understanding its sales performance and customer purchase patterns, which makes it hard to make decisions for inventory, marketing, and planning.

- Which products sell the most and why?
- Which regions achieve high sales?
- Are there specific seasons when sales increase?

Business Requirement

The retail company is interested in obtaining information about its sales and customer trends which may be useful in inventory management and marketing.

This involves determining the top selling products, the performance of each category and region and seasonal trends.

Abstract

The retail sale numbers will be analyzed to get some understanding of the customer buying behaviors and sales trends.

This will recognize the top selling products, best performance regions, and seasonal peaks based on the total sales, product types, regional variations and monthly sales trends.

Results lead to enhanced decision making in inventory management, marketing strategies, and general business planning.

Data Collection

The Dataset contain 500 rows and 7 columns of Retail sales transactions:-

- CustomerID: Unique identifier for each customer
- ProductID: Unique identifier for each product
- Category: Product category
- Quantity: Number of items purchased
- Price: Price per item
- PurchaseDate: Date of purchase
- Region: Customer location

```
> str(data)
Classes 'data.table' and 'data.frame':  500 obs. of  7 variables:
 $ CustomerID  : chr  "C006" "C093" "C023" "C041" ...
 $ ProductID   : chr  "P163" "P769" "P276" "P616" ...
 $ Category    : chr  "Groceries" "Home Appliances" "" "Groceries" ...
 $ Quantity    : num  4 5 3 5 5 2 3 3 3 NA ...
 $ Price       : num  71 605 889 NA 351 406 708 28 186 621 ...
 $ PurchaseDate: IDate, format: "2025-04-11" "2025-01-27" "2025-07-08" "2025-01-15" ...
 $ Region      : chr  "West" "" "East" "West" ...
 - attr(*, ".internal.selfref")=<externalptr>
> summary(data)
  CustomerID         ProductID           Category           Quantity        Price        PurchaseDate           Region
 Length:500         Length:500         Length:500         Min.   :1.00   Min.   : 10.0   Min.   :2025-01-01   Length:500
 Class :character   Class :character   Class :character   1st Qu.:2.00   1st Qu.:207.0   1st Qu.:2025-03-05   Class :character
 Mode  :character   Mode  :character   Mode  :character   Median :3.00   Median :466.0   Median :2025-05-15   Mode  :character
                                                          Mean   :2.96   Mean   :481.3   Mean   :2025-05-12
                                                          3rd Qu.:4.00   3rd Qu.:752.0   3rd Qu.:2025-07-16
                                                          Max.   :5.00   Max.   :999.0   Max.   :2025-09-28
                                                          NA's   :25     NA's   :25
> nrow(data)
[1] 500
```

The dataset is used to analyze sales patterns, top-selling products, sales by category and region, monthly trend.

Load dataset

To load the dataset into R, the data.table package used for its fast reading capabilities.

The fread function reads large datasets efficiently.

```
library(data.table)
data <- fread("D:/BDProject/retail_sales.csv")
```

Data Cleaning

The dataset was checked for missing or inconsistent values, cleaning steps applied using R packages data.table and lubridate.

1. **Handle Missing values:** for the categorical features Region and Category, missing values replaced with “Unknown”.
   And for the numerical features Quantity and Price, missing values replaced with the median to avoid distortion.
2. **Ensure Consistency:** some Products had inconsistent category labels, corrected by assigning the most frequent category per ProductID.
3. **Date format:** The PurchaseDate feature was converted to Date format using the lubridate package.

```r
# check missing values
colSums(is.na(data))
sum(data$Category == "")
sum(data$Region == "")

# Checking for inconsistencies in product categories
inconsistencies_categories <- data[, list(UniqueCategories = uniqueN(Category)), by = ProductID]
products_w_inconsistent <- inconsistencies_categories[UniqueCategories > 1]
products_w_inconsistent

# ensure rows are unique
nrow(unique(data))

# replace missing values, date format and create new feature
retail_data<- data[,
  {
    median_q <- median(Quantity, na.rm = TRUE)
    median_p <- median(Price, na.rm = TRUE)
    Quantity <- ifelse(is.na(Quantity), median_q, Quantity)
    Price <- ifelse(is.na(Price), median_p, Price)

    list(
      CustomerID, ProductID, Quantity, Price,
      Category = ifelse(is.na(Category) | Category == "", "Unknown", Category),
      PurchaseDate = ymd(PurchaseDate),
      Region = ifelse(is.na(Region) | Region == "", "Unknown", Region)
        )
  }
]
#Function to extract the most frequently
get_mode <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}
#Correcting the categories based on the most frequent use of each product.
product_category <- retail_data[, list(CorrectCategory = get_mode(Category)), by = ProductID]
retail_data[product_category, Category := i.CorrectCategory, on = "ProductID"]
#checking after the correcting
inconsistencies_categories2 <- retail_data[, list(UniqueCategories = uniqueN(Category)), by = ProductID]
products_w_inconsistent2 <- inconsistencies_categories2[UniqueCategories > 1]
products_w_inconsistent2
```

```
> colSums(is.na(data))
  CustomerID    ProductID     Category     Quantity        Price PurchaseDate       Region
           0            0            0           25           25            0            0
> sum(data$Category == "")
[1] 25
> sum(data$Region == "")
[1] 25
> # ensure rows are unique
> nrow(unique(data))
[1] 500
> products_w_inconsistent2
Empty data.table (0 rows and 2 cols): ProductID,UniqueCategories
> summary(retail_data)
  CustomerID         ProductID           Quantity         Price          Category         PurchaseDate          Region
 Length:500         Length:500         Min.   :1.000   Min.   : 10.0   Length:500         Min.   :2025-01-01   Length:500
 Class :character   Class :character   1st Qu.:2.000   1st Qu.:222.8   Class :character   1st Qu.:2025-03-05   Class :character
 Mode  :character   Mode  :character   Median :3.000   Median :466.0   Mode  :character   Median :2025-05-15   Mode  :character
                                       Mean   :2.962   Mean   :480.5                      Mean   :2025-05-12
                                       3rd Qu.:4.000   3rd Qu.:741.2                      3rd Qu.:2025-07-16
                                       Max.   :5.000   Max.   :999.0                      Max.   :2025-09-28
> colSums(is.na(retail_data))
  CustomerID    ProductID     Quantity        Price     Category PurchaseDate       Region
           0            0            0            0            0            0            0
> sum(c(retail_data$Category,retail_data$Region) == "")
[1] 0
```

```
> str(retail_data)
Classes ‘data.table’ and ‘data.frame’:  500 obs. of  8 variables:
 $ CustomerID  : chr  "C006" "C093" "C023" "C041" ...
 $ ProductID   : chr  "P163" "P769" "P276" "P616" ...
 $ Quantity    : num  4 5 3 5 5 2 3 3 3 3 ...
 $ Price       : num  71 605 889 466 351 406 708 28 186 621 ...
 $ Category    : chr  "Groceries" "Home Appliances" "Unknown" "Groceries" ...
 $ PurchaseDate: Date, format: "2025-04-11" "2025-01-27" "2025-07-08" "2025-01-15" ...
 $ Region      : chr  "West" "Unknown" "East" "West" ...
 $ TotalSales  : num  284 3025 2667 2330 1755 ...
 - attr(*, ".internal.selfref")=<externalptr>
> head(retail_data)
   CustomerID ProductID Quantity Price        Category PurchaseDate  Region TotalSales
       <char>    <char>    <num> <num>          <char>       <Date>  <char>      <num>
1:       C006      P163        4    71       Groceries   2025-04-11    West        284
2:       C093      P769        5   605 Home Appliances   2025-01-27 Unknown       3025
3:       C023      P276        3   889         Unknown   2025-07-08    East       2667
4:       C041      P616        5   466       Groceries   2025-01-15    West       2330
5:       C029      P351        5   351            Toys   2025-04-22    West       1755
6:       C098      P448        2   406 Home Appliances   2025-05-11   South        812
```

Compute Total Sales

A new feature TotalSales was created by multiplying Quantity by Price for each row.

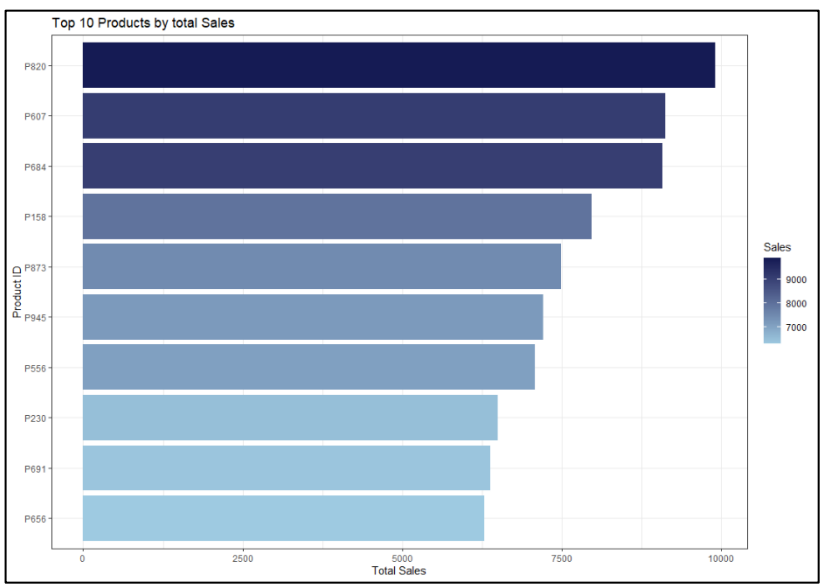
this feature can be used in analyses like identifying top products, and category sales.

```
retail_data[,TotalSales:=Quantity*Price]
```

```
> str(retail_data)
Classes 'data.table' and 'data.frame':  500 obs. of  8 variables:
 $ CustomerID  : chr  "C006" "C093" "C023" "C041" ...
 $ ProductID   : chr  "P163" "P769" "P276" "P616" ...
 $ Quantity    : num  4 5 3 5 5 2 3 3 3 3 ...
 $ Price       : num  71 605 889 466 351 406 708 28 186 621 ...
 $ Category    : chr  "Groceries" "Home Appliances" "Unknown" "Groceries" ...
 $ PurchaseDate: Date, format: "2025-04-11" "2025-01-27" "2025-07-08" "2025-01-15" ...
 $ Region      : chr  "West" "Unknown" "East" "West" ...
 $ TotalSales  : num  284 3025 2667 2330 1755 ...
 - attr(*, ".internal.selfref")=<externalptr>
> head(retail_data)
   CustomerID ProductID Quantity Price        Category PurchaseDate  Region TotalSales
       <char>    <char>    <num> <num>          <char>       <Date>  <char>       <num>
1:       C006      P163        4    71       Groceries   2025-04-11    West         284
2:       C093      P769        5   605 Home Appliances   2025-01-27 Unknown        3025
3:       C023      P276        3   889         Unknown   2025-07-08    East        2667
4:       C041      P616        5   466       Groceries   2025-01-15    West        2330
5:       C029      P351        5   351            Toys   2025-04-22    West        1755
6:       C098      P448        2   406 Home Appliances   2025-05-11   South         812
```

Top Product Analysis

The top 10 products by total sales were identified using the TotalSales feature that helps to understand which products make the highest sales and should be most priority in the retail strategy.




```
top_products <- retail_data[
  , list(Total_Sales = sum(TotalSales)), by = ProductID
][
  order(desc(Total_Sales))
][
  1:10
]

top_products

ggplot(top_products,
  aes(x = reorder(ProductID, Total_Sales), y = Total_Sales,fill = Total_Sales)) +
  geom_col() +
  scale_fill_gradient(low = "#9ecae1", high = "#151B54", name = "Sales")+
  coord_flip() +
  ggtitle("Top 10 Products by total Sales") +
  xlab("Product ID") +
  ylab("Total Sales") +
  theme_bw()
```

```
> top_products
    ProductID Total_Sales
       <char>       <num>
 1:      P820        9908
 2:      P607        9118
 3:      P684        9082
 4:      P158        7965
 5:      P873        7495
 6:      P945        7205
 7:      P556        7078
 8:      P230        6501
 9:      P691        6378
10:      P656        6291
```

Product P820 achieve high sales with a total of 9908, indicate its strong demand among customers.

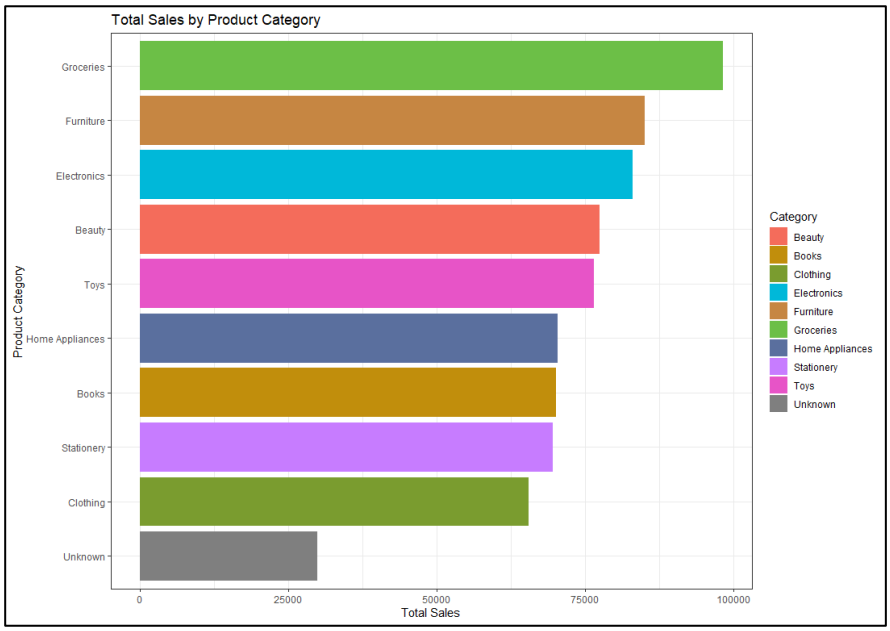
Sales by Category

The total sales were grouped by product category, that revealing the top selling categories and showing how sales are distributed in different categories.

```
Category_colors <- c( "Beauty" = "#F46C5B","Books" = "#C18E0C","Clothing" = "#7A9C2F","Electronics" = "#00B8D9"
                      ,"Furniture" = "#C68642","Groceries" = "#6CBF47","Home Appliances" = "#5A6F9E","Stationery" = "#C77CFF","Toys" = "#E754C7","Unknown"="#7f7f7f")

sales_by_category <- retail_data[ ,list(Total_Sales = sum(TotalSales)), by = Category ]

ggplot(sales_by_category,
  aes(x = reorder(Category, Total_Sales), y = Total_Sales,fill = Category)) +
  geom_col() +
  scale_fill_manual(values = Category_colors)+
  coord_flip() +
  ggtitle("Total Sales by Product Category")+
  xlab("Product Category")+
  ylab("Total Sales") +
  theme_bw()
```




```
> sales_by_category
           Category Total_Sales
             <char>       <num>
 1:        Groceries       98206
 2: Home Appliances       70389
 3:          Unknown       29837
 4:             Toys       76575
 5:           Beauty       77458
 6:            Books       70116
 7:         Clothing       65524
 8:        Furniture       85033
 9:       Stationery       69628
10:      Electronics       83018
```

The Groceries category had the highest sales of 79102, while the Home Appliances category had the lowest at 55175.

Regional Sales Analysis

Sales were analyzed by region and product category to identify high performing areas and find the distribution of sales across categories and determine each region's share of total sales.

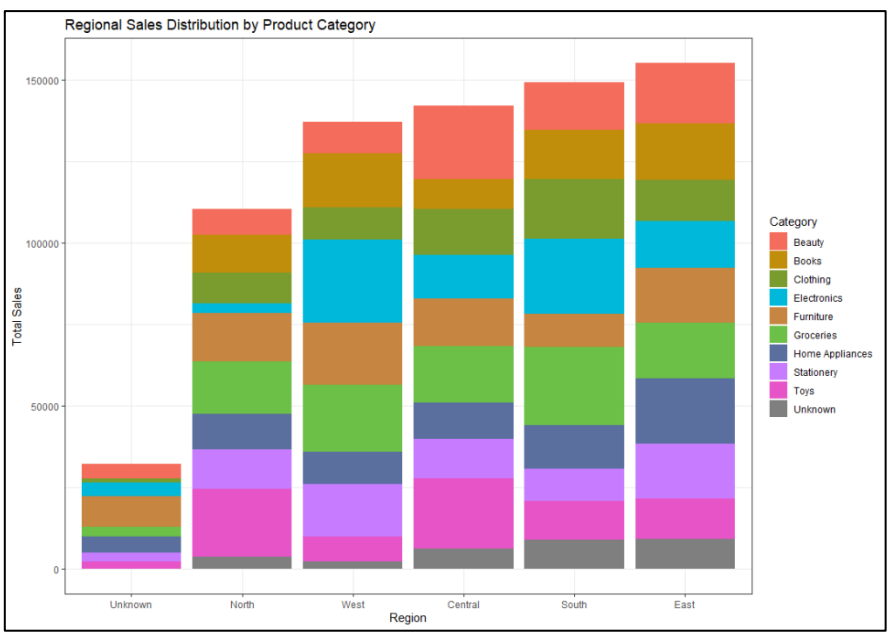
```
regional_sales_category <- retail_data[,list(Total_Sales = sum(TotalSales)), by = list(Region, Category)]

ggplot(regional_sales_category,
  aes(x = reorder(Region, Total_Sales), y = Total_Sales,fill = Category)) +
  geom_col() +
  scale_fill_manual(values = Category_colors)+
  ggtitle("Regional Sales Distribution by Product Category")+
  xlab("Region")+
  ylab("Total Sales")+
  theme_bw()

region_distribution <- retail_data[, list(Total_Sales = sum(TotalSales)), by = Region]
region_distribution[, Percentage := round((Total_Sales / sum(Total_Sales)) * 100, 1)]

ggplot(region_distribution,
  aes(x = "", y = Total_Sales, fill = Region)) +
  geom_col(color = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = paste0(Region, "\n", Percentage, "%")),
            position = position_stack(vjust = 0.5),
            color = "white", size = 5) +
  ggtitle("Sales Distribution by Region")+
  theme_void()
```

Regional Sales by Category

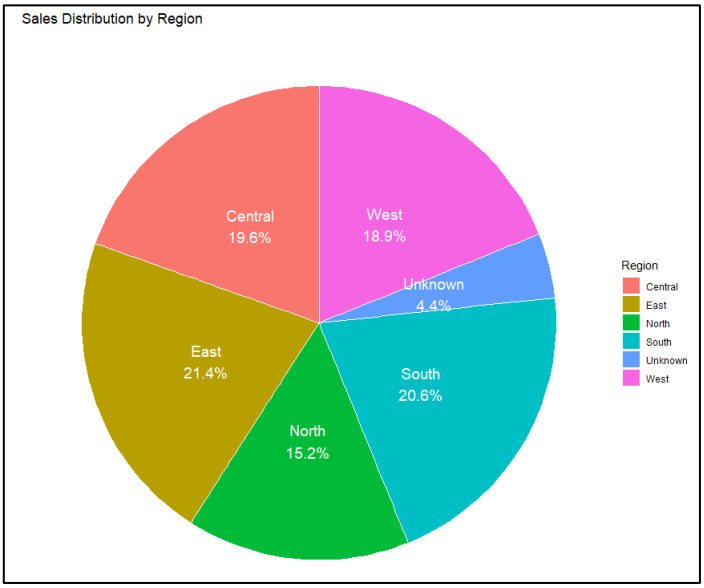


The chart showing total sales by region and product category shows clear regional differences.

- **The East region** achieved highest overall sales, mostly led by Beauty, Books, and Home Appliances categories.
- **The South region** Ranked second in overall sales, led by Groceries and Electronics.
- **The North and Central regions** show high sales in Toys and Books, reflecting customer interest in entertainment and educational products.
- **The west region** had a good performance, with its diverse sales led by Electronics, Groceries, and Stationery.

This indicates that consumer preferences vary across regions.

Sales distribution by Region




```
> region_distribution
    Region Total_Sales Percentage
    <char>       <num>      <num>
1:    West      137052       18.9
2: Unknown       32056        4.4
3:    East      155066       21.4
4:   South      149343       20.6
5:   North      110275       15.2
6: Central      141992       19.6
```

The chart reveals the percentage of total sales made by each region.

- **East** has the highest share of total sales.
- The **South**, **Central**, and **West** regions have relatively similar shares.
- **North** has the lowest share.
- **Unknown** represents sales where the Regions location has not been defined.

Overall, Sales are relatively balanced across regions.

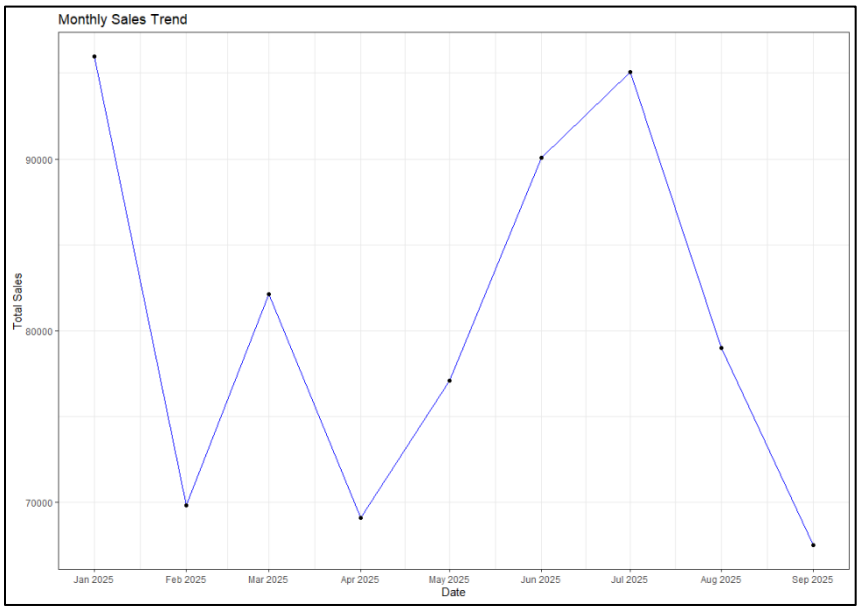
Monthly Sales Trend

Sales over time are analyzed to identify trend and seasonal peak months of the year.

This indicates which months are the biggest contribution to the total sales.

```
monthly_sales <- retail_data[,list(Total_Sales = sum(TotalSales)), by = list(YearMonth = floor_date(PurchaseDate, "month")) ][
  order(YearMonth)
  ]

ggplot(monthly_sales,
  aes(x = YearMonth, y = Total_Sales)) +
  geom_line(color = "blue") +
  geom_point() +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %Y") +
  ggtitle("Monthly Sales Trend")+
  xlab("Date")+
  ylab("Total Sales")+
  theme_bw()
```




```
> monthly_sales
    YearMonth Total_Sales
       <Date>       <num>
1: 2025-01-01       95953
2: 2025-02-01       69830
3: 2025-03-01       82133
4: 2025-04-01       69114
5: 2025-05-01       77109
6: 2025-06-01       90062
7: 2025-07-01       95046
8: 2025-08-01       79004
9: 2025-09-01       67533
```

Peak months in sales are January and July, and these months reflect peak customer activity. While September recorded the lowest sales, And Sales saw general trend growth start from from April to July then dropped in August.

Overall, there is a downward trend in sales in the first half of the year, followed by gradual increase in sales, that peaks almost in the middle of the year.

Conclusion

analysis of retail sales data provides insights into customer purchase behavior and the performance of sales across products, categories, and regions.

Key Findings:

- Top selling Product is P820, with Total sales of 9908, showing strong customer demand.
- Top Categories is Groceries, which achieve total sales of 79102, while Home Appliances had the lowest total sales. This confirms the primary revenue streams and areas that require strategic focus.
- Regional analysis shows that the East with 21.4% and South with 20.6%, contribute the most to total sales, placing them at the top of the market. while preferences difference across product categories like Beauty in the East and Groceries in the West.
- Seasonal Trends reveal peak sales during January and July show reflect seasonal customer activity. This pattern demands careful planning of inventory and marketing campaigns around these months.

These insights will help the retail company in the effective management of inventory, planning of marketing strategies in a region, and take decisions that ensure maximization of revenue by meeting customer needs efficiently.